Higgs Couplings at LhmuC

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- Higgs XS and production rates

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(l\,p \to \nu_l\,H + \text{jet} + \text{X})$ [fb]	52	164	438	731
$\sigma(l\,p \to l\,H + \text{jet} + \text{X})$ [fb]	5.6	19	135	221
N_ev (CC)		**164K**	**109.5K**	**365K**

What if we do a simple rescaling?

- LheC Higgs Signal Strenght projections

Setup	$b\bar{b}$	$b\bar{b} \oplus$ Thy	WW	gg	$\tau\tau$	cc	ZZ	$\gamma\gamma$
LHeC NC	2.3	2.4	17	16	15	20	35	42
LHeC CC	0.80	0.94	6.2	5.8	5.2	7.1	12	15

[2007.14491]

X(0.82)1.29 sensitivity

X(1.5)1.8 sensitivity

(Not) Assuming LheC baseline

Is this a good approximation?

- Compare with Simon and Kingman [2101.10476 +WP]

$$\frac{\Delta g_{hbb}}{g_{hbb}}$$

500 GeV: 0.58%

1 TeV: 0.52%

LheC: 0.97%

Assuming **NO** LheC baseline

Disagreement between naive expectations and results

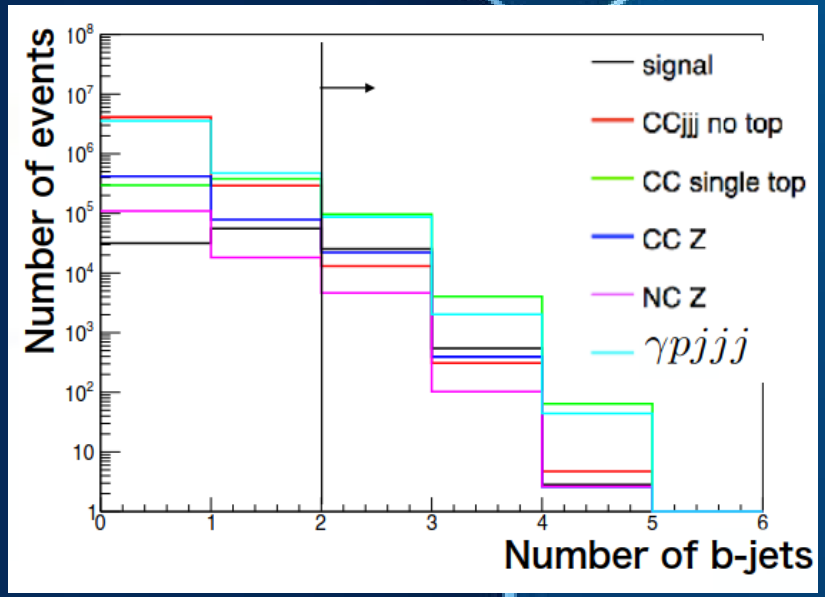
Possible reason?

- Compare with Simon and Kingman [2101.10476 +WP]

Dominant background in [4]: WW -> Z-boson production with Z → bb.

- For LheC however the highest bckg cited is CC > jjj

QUESTION?
has this been considered in the bb ongoing studies (anomalous H couplings, Higgs measurements) at LhmuC?



Naive estimations

- generate mu+ p > j j j vm~ (minimal MG cuts on ptj, eta)
  - @LheC(P = -80, 60 GeV) = 11.34 pb
  - @LhmuC(P= 0, 500 GeV) = 34.8 pb

- generate mu+ p > z j vm~, z > b b~ (minimal MG cuts on ptj, eta)
  - @LheC(P = -80, 60 GeV) = 0.137pb
  - @LhmuC(P= 0, 500 GeV) = 0.293pb

With the same minimal cuts
CC Z = 0.8(1.2) % CC jjj @ LhmuC (LHeC)

Remarks

- Naive rescaling of LheC SS seems to fail, 2 possible explanations (as far as I can think of) :
    - Relevant background has been neglected in previous studies?
    - The background was considered but at LhμC the signal is much better identified?

FIRST CASE: LheC SS can be recycled if after including the background the naive expectations turn realistic

SECOND CASE: LheC SS cannot be recycled and more studies are needed to perform a Higgs couplings fit